EXPERIMENT : 15

Design and Simulation of an IoT-Based Smart Agriculture System

OBJECTIVES :

1. To understand the architecture and working principles of an IoT-based smart agriculture system.

2. To simulate the collection of soil moisture, temperature, and humidity data using MATLAB.

3. To design an automatic irrigation control algorithm based on soil moisture threshold values.

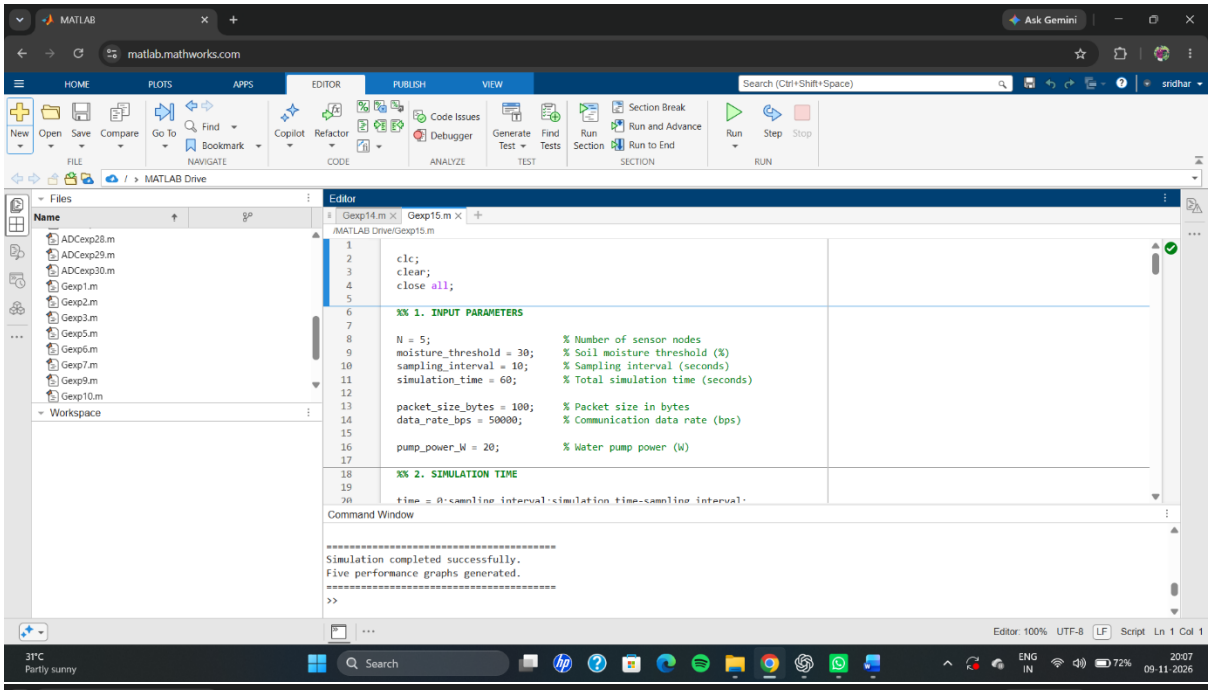
4. To calculate the number of sensor data packets, total transmitted data, and communication time.

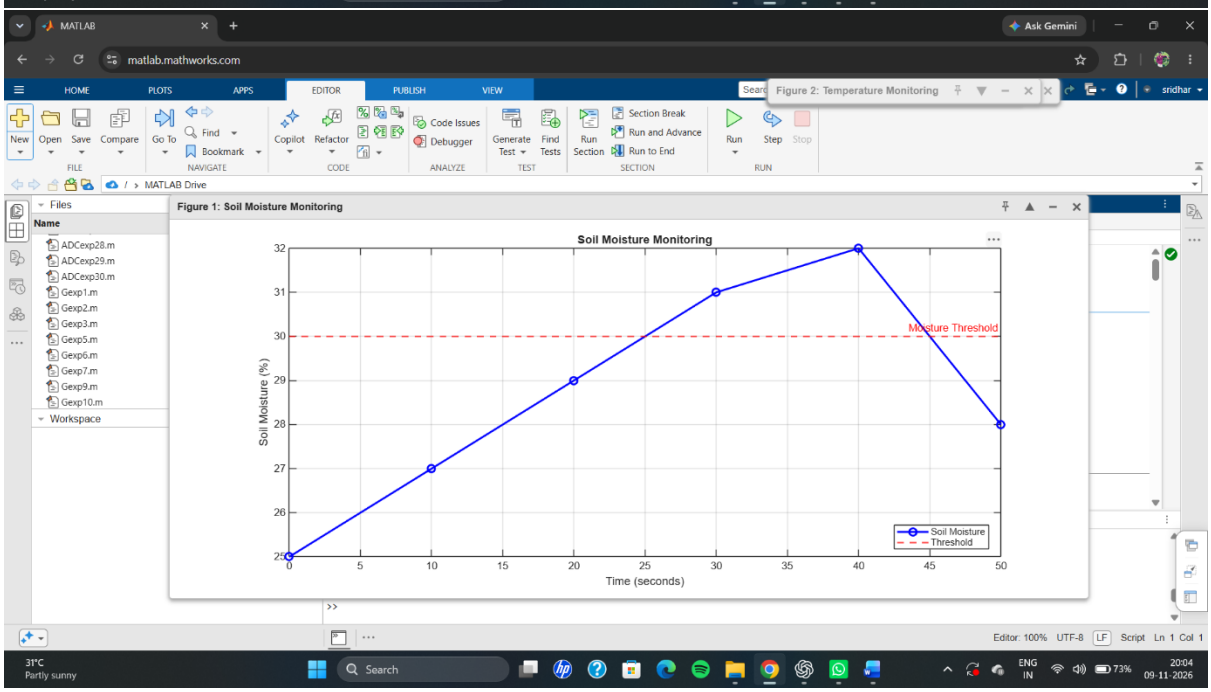
5. To estimate water pump energy consumption based on its operating duration and power rating.

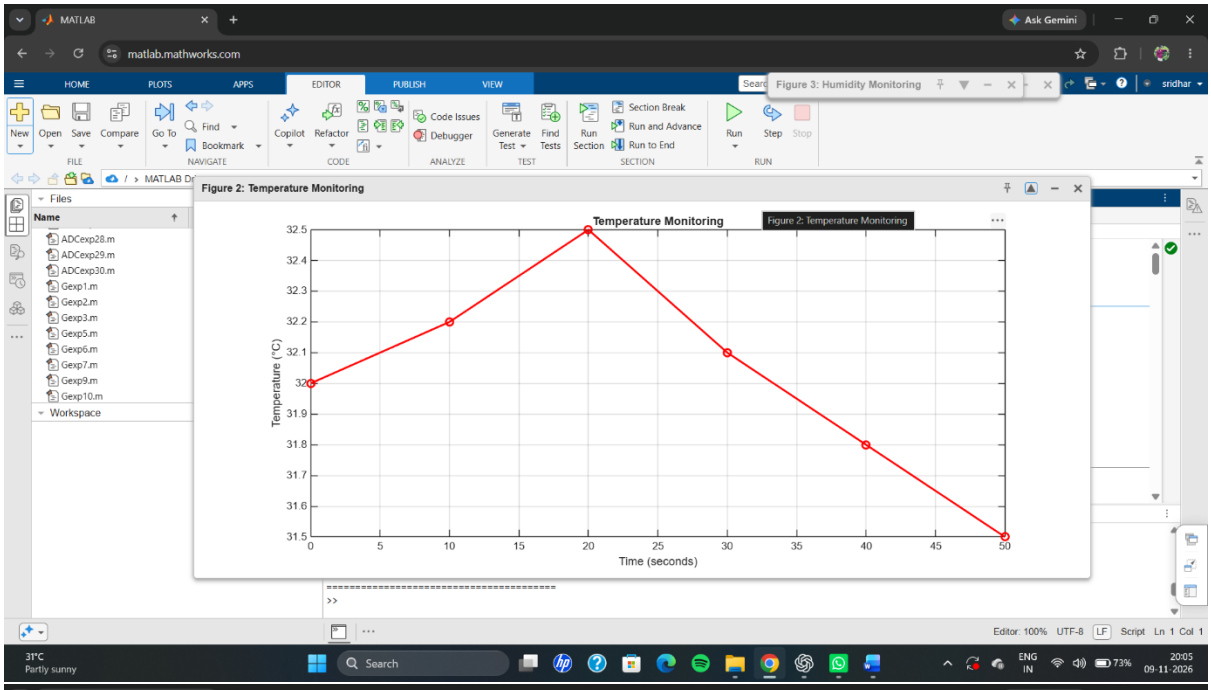
6. To generate and analyze graphs representing environmental parameters and irrigation pump status.

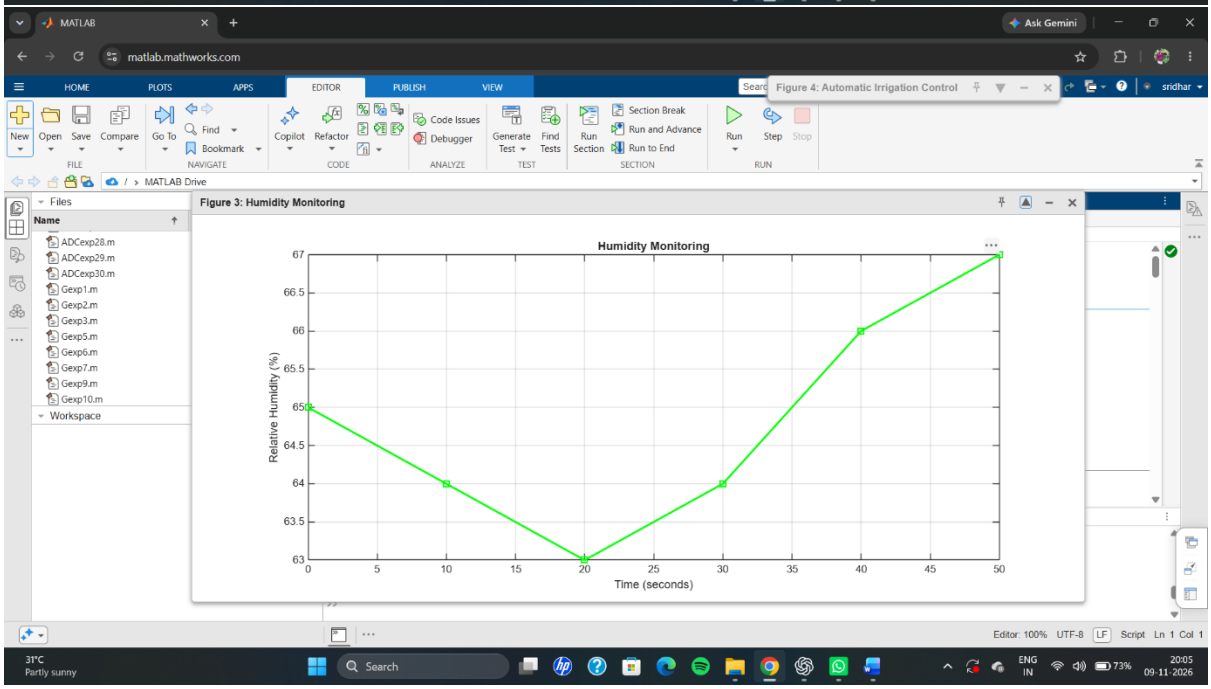
7. To evaluate how IoT technology can improve water management and agricultural monitoring.

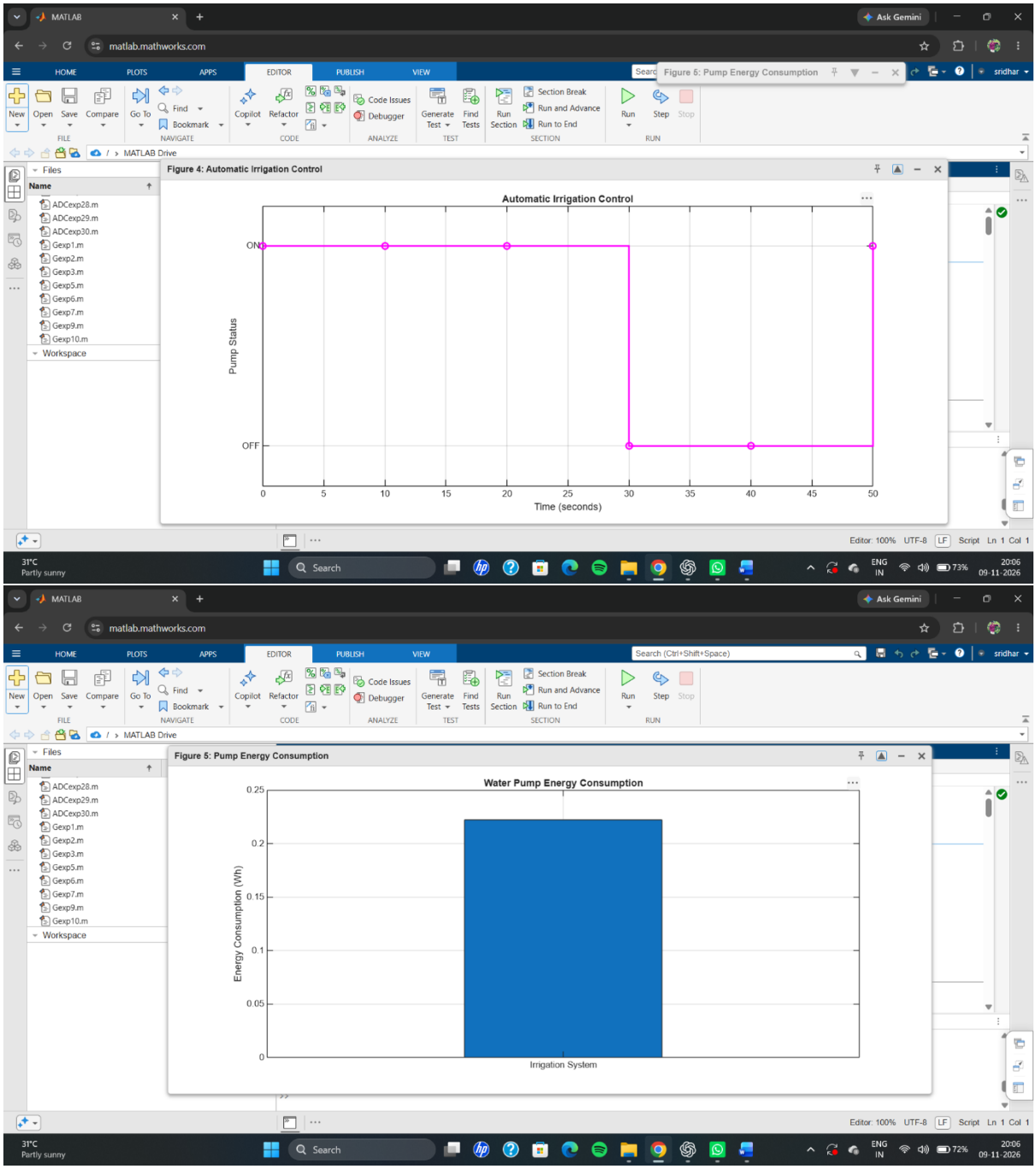
PROGRAME CODE :


MATLAB
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
Search (Ctrl+Shift+Space)
sridhar
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Files
Name
ADCexp28.m
ADCexp29.m
ADCexp30.m
Gexp1.m
Gexp2.m
Gexp3.m
Gexp5.m
Gexp6.m
Gexp7.m
Gexp9.m
Gexp10.m
Workspace
Editor
Gexp14.m
Gexp15.m
/MATLAB Drive/Gexp15.m
clc;
clear;
close all;
%% 1. INPUT PARAMETERS
N = 5; % Number of sensor nodes
moisture_threshold = 30; % Soil moisture threshold (%)
sampling_interval = 10; % Sampling interval (seconds)
simulation_time = 60; % Total simulation time (seconds)
packet_size_bytes = 100; % Packet size in bytes
data_rate_bps = 50000; % Communication data rate (bps)
pump_power_W = 20; % Water pump power (W)
%% 2. SIMULATION TIME
time = 0:sampling_interval:simulation_time-sampling_interval;
Command Window
Simulation completed successfully.
Five performance graphs generated.
>>
Editor: 100% UTF-8 LF Script Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
72%
20:07
09-11-2026


MATLAB
Ask Gemini
matlab.mathworks.com
Figure 2: Temperature Monitoring
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
sridhar
MATLAB Drive
Files
Name
ADCexp28.m
ADCexp29.m
ADCexp30.m
Gexp1.m
Gexp2.m
Gexp3.m
Gexp5.m
Gexp6.m
Gexp7.m
Gexp9.m
Gexp10.m
Workspace
Figure 1: Soil Moisture Monitoring
Soil Moisture Monitoring
Moisture Threshold
Soil Moisture (%)
Time (seconds)
Soil Moisture
Threshold
32
31
30
29
28
27
26
25
0
5
10
15
20
25
30
35
40
45
50
>>
Editor: 100% UTF-8 LF Script Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
73%
20:04
09-11-2026


Figure 2: Temperature Monitoring
Temperature Monitoring
Temperature (°C)
Time (seconds)


Figure 3: Humidity Monitoring
Humidity Monitoring
Relative Humidity (%)
Time (seconds)



RESULT : The MATLAB simulation of the IoT-based Smart Agriculture and Automatic Irrigation System was successfully implemented. Soil moisture, temperature, and humidity were monitored, and the water pump was automatically controlled based on the predefined soil moisture threshold of 30%. The system operated the pump for 40 seconds and consumed 800 J of energy. The simulation demonstrates how IoT-based monitoring and automatic irrigation can reduce unnecessary water usage and support efficient agricultural management.